

TITLE PAGE

- **Problem Statement ID – Question 1**
- **Problem Statement Title-**AI-Powered Multi-Agent Building Energy Optimization
using Large Language Models
- **Theme-** Smart Automation / Artificial Intelligence / Sustainable Buildings
- **PS Category-** Software
- **Student Name-** Pallavi
- **Student ID-**22MIP10086

IDEA TITLE

Proposed Solution:- [EcoLoop : AI-Powered Multi-Agent Building Energy Optimization System.](#)

Explanation:- EcoLoop is an intelligent building management system that combines **EnergyPlus building simulation**, **Large Language Models (Qwen)**, and **AI agents** to optimize HVAC operations in real time.

The system continuously monitors:- **Indoor temperature, Humidity, Occupancy, Building energy consumption.**

The LLM analyzes these inputs and generates intelligent decisions such as:

- Adjusting cooling setpoints
- Optimizing fan speed
- Reducing unnecessary lighting
- Improving occupant comfort
- These decisions reduce electricity consumption while maintaining indoor thermal comfort.

❑ **How it addresses the problem**

Reduces unnecessary HVAC usage

Lowers electricity consumption

Maintains occupant comfort

Supports sustainable building operation

Enables autonomous decision making

Innovation & Uniqueness

LLM-powered autonomous building management

Multi-agent architecture

Closed-loop optimization

EnergyPlus integration

AI-driven HVAC optimization

Real-time decision making

TECHNICAL APPROACH

.Python

- Google Colab

- Qwen LLM

- EnergyPlus

- Pandas

- NumPy

- Matplotlib

- JSON

- GitHub

FEASIBILITY AND VIABILITY

❑ Feasibility

- Uses open-source tools
- Easily deployable
- Low implementation cost
- Compatible with existing buildings
- Scalable for offices, schools, hospitals, and smart buildings

❑ Potential Challenges

- Integration with EnergyPlus
- Runtime optimization
- Accurate sensor data
- LLM response latency
- Large simulation time

❑ Mitigation

- Efficient prompt engineering
- Optimized simulation loops
- Real-time monitoring
- EnergyPlus Python API
- Modular software architecture

ARTIFACTS

```
Eco-Loop Building Agents
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[3] 1m
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

model_name = "Qwen/Qwen2.5-1.5B-Instruct"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float16,
    device_map="auto"
)

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
WARNING: huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
config.json: 100% ██████████ 660/660 [00:00<00:00, 59.5kB/s]

tokenizer_config.json: 100% ██████████ 7.30k/7.30k [00:00<00:00, 298kB/s]

vocab.json: 100% ██████████ 2.78M/2.78M [00:00<00:00, 25.8MB/s]

merges.txt: 100% ██████████ 1.67M/1.67M [00:00<00:00, 47.8MB/s]

tokenizer.json: 100% ██████████ 7.03M/7.03M [00:00<00:00, 62.5MB/s]
[transformers] `torch_dtype` is deprecated! Use `dtype` instead!

model.safetensors: reconstructing file: 100% ██████████ 3.09GB / 3.09GB, 87.2MB/s

model.safetensors: downloading bytes: ██████████ 2.69GB, 151MB/s

Loading weights: 100% ██████████ 338/338 [00:06<00:00, 30.96it/s]

generation_config.json: 100% ██████████ 242/242 [00:00<00:00, 24.2kB/s]
```

```
Eco-Loop Building Agents
File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Time Temperature Electricity PMV
0 0 23 185 0.8
1 1 23 185 0.8
2 2 23 185 0.8
3 3 23 185 0.8
4 4 23 185 0.8

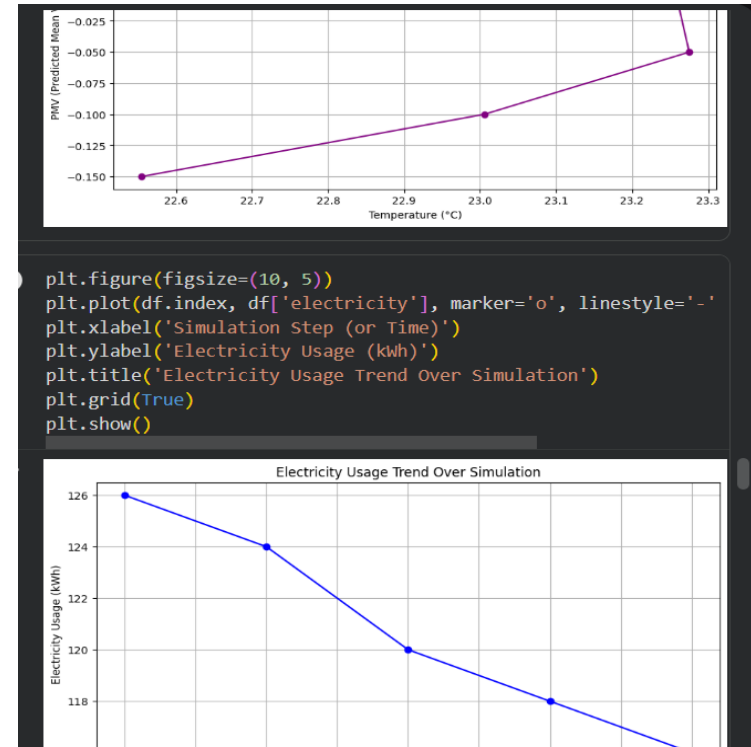
Next steps: Generate code with df New interactive sheet

[53] 3s
from google.colab import sheets
sheet = sheets.InteractiveSheet(df=df)

https://docs.google.com/spreadsheets/d/1H1vf3oQPtmEY2o2dYgKT7JK

File Edit View Insert Format Data Tools Extensions H

A1 temperature
A B C D
1 temperature humidity occupancy electricity pmv
2 22.78905903 66.9630585 120 126
3 23.2563445 65.10844461 120 124
4 23.27509893 66.20807198 120 120
5 23.00528839 65.65835715 120 118
6 22.55402823 65.26597743 120 116
```



```
from google.colab import drive
drive.mount('/content/drive')

Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_remount=True).

import os

BASELINE_IDF = "/content/drive/MyDrive/EcoLoop_Project/business_model /baseline_building.idf"
OPTIMIZED_IDF = "/content/drive/MyDrive/EcoLoop_Project/business_model /optimized_building.idf"

print("Baseline File:", BASELINE_IDF)
print("Optimized File:", OPTIMIZED_IDF)

Baseline File: /content/drive/MyDrive/EcoLoop_Project/business_model /baseline_building.idf
Optimized File: /content/drive/MyDrive/EcoLoop_Project/business_model /optimized_building.idf

print("Baseline Exists:", os.path.exists(BASELINE_IDF))
print("Optimized Exists:", os.path.exists(OPTIMIZED_IDF))

Baseline Exists: True
Optimized Exists: True
```

RESEARCH AND REFERENCES

- EnergyPlus Documentation
- EnergyPlus Python API
- Qwen LLM Documentation
- Hugging Face Transformers
- Python Documentation
- Pandas Documentation
- GitHub
- ASHRAE Standards for Thermal Comfort